Boundary Conditions for Electromagnetics

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

IRA A. FULTON SCHOOLS OF engineering
ARIZONA STATE UNIVERSITY

Lecture Objective

- To demonstrate the conditions that electromagnetic fields and electric current must obey at material boundaries.

Boundary Conditions

- The behavior of the electromagnetic field at the boundary between two different media can be derived from Maxwell's equations in integral form.
- Boundary conditions specify what happens to the normal and tangential components of the field at the abrupt change from one medium to the other.
- To derive the boundary conditions, we apply the integral form of the fundamental laws to an infinitesimally small region that lies partially in one medium and partially in the other.

Maxwell's Equations in Integral Form

$$\int_S \vec{D} \cdot d\vec{s} = \int_V \rho_V dv$$

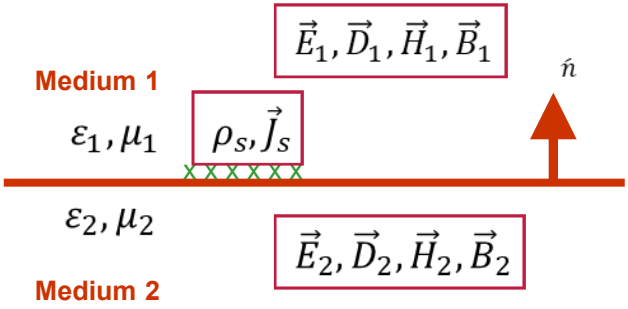
$$\int_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{s}$$

$$\int_S \vec{B} \cdot d\vec{s} = 0$$

$$\int_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{s} + \frac{d}{dt} \int_S \vec{D} \cdot d\vec{s}$$

Derivation of Boundary Conditions

- Consider two semi-infinite media separated by a boundary. A surface charge and/or surface current may exist at the interface.
- Locally, the boundary will look planar



Boundary Conditions on Flux Densities

- Recall that if a vector field is solenoidal, then its normal component is continuous across the material boundary.
- Hence, the boundary condition on the magnetic flux density is

$$\boxed{B_{1n} = B_{2n}} \quad \Rightarrow \quad \boxed{\hat{n} \cdot \left(\vec{B}_1 - \vec{B}_2\right) = 0}$$

Boundary Conditions on Flux Densities continued

- If the charge density is zero, then the normal component of displacement flux density is also continuous across the boundary.
- Furthermore, the total charge enclosed by an infinitesimal volume will vanish unless the charge occurs as a surface charge density.
- We find that

$$\boxed{D_{1n} - D_{2n} = \rho_s} \quad \Rightarrow \quad \boxed{\hat{n} \cdot \left(\vec{D}_1 - \vec{D}_2\right) = 0}$$

Boundary Conditions on Field Intensities

- Recall that if a vector field is irrotational, then its tangential component is continuous across the material boundary.
- Hence, the boundary condition on the electric field intensity is

$$\boxed{E_{1t} = E_{2t}} \quad \Rightarrow \quad \boxed{\hat{n} \times \left(\vec{E}_1 - \vec{E}_2\right) = 0}$$

Boundary Conditions on Field Intensities continued

- If the current density is zero, then the tangential component of magnetic field intensity is also continuous across the boundary.
- Furthermore, the total current flowing through an infinitesimal surface will vanish unless the current occurs as a surface current density.
- We find that

$$\boxed{\hat{n} \times \left(\vec{H}_1 - \vec{H}_2\right) = \vec{J}_s}$$

Electric Field Boundary Conditions - Special Cases

- Special Case 1: the interface between two perfect (non-conducting) dielectrics:
  - Physical principle: “there can be no free surface charge associated with the surface of a perfect dielectric.”
  - In practice: unless an impressed surface charge is explicitly stated, assume it is zero.

$$\boxed{\begin{aligned} D_{1n} &= D_{2n} \\ E_{1t} &= E_{2t} \end{aligned}}$$

Electric Field Boundary Conditions - Special Cases continued

- Special Case 2: *the interface between a conductor and a perfect dielectric*:
  - Physical principle: "there can be no electrostatic field inside of a conductor."
  - In practice: a surface charge always exists at the boundary.

$$D_{1n} = \rho_s$$
$$E_{1t} = 0$$

Magnetic Field Boundary Conditions – Special Case

- In a perfect conductor, both the electric and magnetic fields must vanish in its interior. Thus,

$$B_n = 0$$

$$\hat{n} \times \overline{H} = \overline{J}_s$$

- a surface current must exist
- the magnetic field just outside the perfect conductor must be tangential to it.

Boundary Conditions for Steady Current Flow

- The behavior of current flow across the interface between two different materials is governed by boundary conditions.
- The boundary conditions for current flow are obtained from the *integral forms* of the basic equations governing current flow.

Boundary Conditions for Steady Current Flow continued

- The governing equations for <u>steady</u> electric current (in a conductor) are:

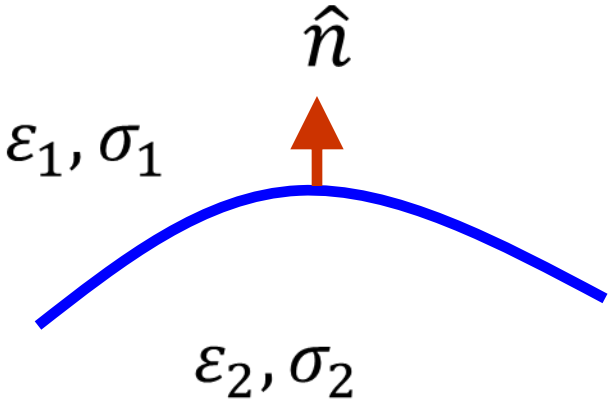
$$\oint_S \bar{J} \cdot d\bar{s} = 0$$

The current density is a solenoidal vector.

$$\oint_C \bar{E} \cdot d\bar{l} = 0 \quad \Rightarrow \quad \oint_C \frac{\bar{J}}{\sigma} \cdot d\bar{l} = 0$$

The ratio of current density to conductivity is a conservative vector.

Boundary Conditions for Steady Current Flow Continues



Boundary Conditions for Steady Current Flow continued 2

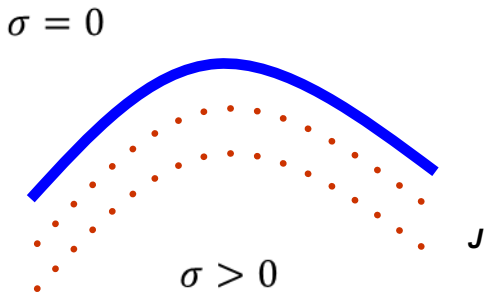
- The normal component of a ***solenoidal*** vector field is continuous across a material interface:

$$J_{1n} = J_{2n}$$

- The tangential component of a ***conservative*** vector field is continuous across a material interface:

$$\frac{J_{1t}}{\sigma_1} = \frac{J_{2t}}{\sigma_2}$$

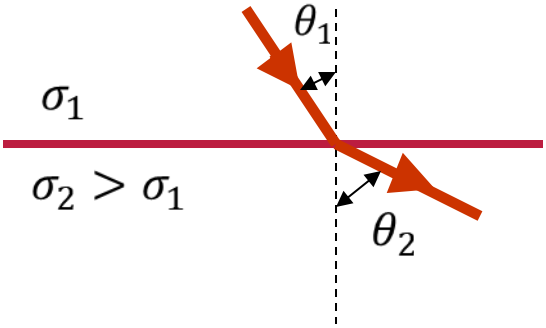
Conductor-Dielectric Interface



- The current in the conductor must flow tangential to the boundary surface.
- The tangential component of the electric field must be continuous across the interface.
- The normal component of the electric field must be zero at the boundary inside the conductor, but not in the insulator. Thus, there will be a buildup of surface charge at the interface.

Conductor-Conductor Interface continued

- The lines of current flow bend as they cross the interface between two conductors



Conductor-Conductor Interface

- The angles are related by $\theta_2 = \tan^{-1}\left(\frac{\sigma_2}{\sigma_1}\tan\theta_1\right)$

- Suppose medium 1 is a good conductor and medium 2 is a good insulator (i.e., $\sigma_1 >> \sigma_2$). Then $\theta_2 \rightarrow 0$. In other words, the current enters medium 2 at nearly right angles to the boundary. This result is consistent with the fact that the electric field in medium 2 should have a vanishingly small tangential component at the interface.

Question

- What happens if two wires, one made from aluminum and the other from copper, are connected together colinearly?
- Assume the resulting wire is carrying a current of 10 (A), has a circular cross section of radius 10 (mm), and the interface between aluminum and copper is a perfectly smooth circular junction. What is the surface charge density at the junction?

Summary

- Boundary conditions on electromagnetic fields can be derived from the integral form of Maxwell's equations.
- We have examined the boundary conditions in general and for several special cases.

Ira A. Fulton Schools of Engineering

Arizona State University